Advanced Coordinate Geometry

ANSWER KEY

Section 1: Ellipses & Focal Properties

1 What are the coordinates of the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$?

a. $(\pm 4, 0)$ ✓

b. $(0, \pm 4)$

c. $(\pm 5, 0)$

d. $(0, \pm 3)$

Since $a^2 = 25 > b^2 = 9$, the major axis is horizontal. $c^2 = a^2 - b^2 = 25 - 9 = 16 \Rightarrow c = 4$. Foci are at $(\pm 4, 0)$.

2 What is the length of the minor axis of the ellipse $\frac{(x-2)^2}{16} + \frac{(y+3)^2}{36} = 1$?

a. 4

b. 8 ✓

c. 6

d. 12

$b^2 = 16 \Rightarrow b = 4$. The length of the minor axis is $2b = 2(4) = 8$.

3 What is the eccentricity of the ellipse $9x^2 + 25y^2 = 225$?

a. $\frac{4}{5}$ ✓

b. $\frac{3}{5}$

c. $\frac{16}{25}$

d. $\frac{5}{4}$

Divide by 225: $\frac{x^2}{25} + \frac{y^2}{9} = 1 \Rightarrow a = 5, b = 3, c = \sqrt{25 - 9} = 4$. Eccentricity $e = \frac{c}{a} = \frac{4}{5}$.

Section 2: Hyperbolas & Asymptotes

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4 What are the equations of the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$?

a. $y = \pm\frac{3}{4}x$ ✓

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

For a horizontal hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the asymptotes are $y = \pm\frac{b}{a}x$. Here $a = 4$ and $b = 3$, so $y = \pm\frac{3}{4}x$.

5 What is the distance between the two vertices of the hyperbola $\frac{y^2}{64} - \frac{x^2}{36} = 1$?

a. 12

b. 16 ✓

c. 20

d. 24

The transverse axis is vertical along y. $a^2 = 64 \Rightarrow a = 8$. The distance between vertices is $2a = 2(8) = 16$.

6 What is the focal length (distance from center to a focus) of $\frac{x^2}{144} - \frac{y^2}{25} = 1$?

a. 11

b. 12

c. 13 ✓

d. 17

For a hyperbola, $c^2 = a^2 + b^2 = 144 + 25 = 169 \Rightarrow c = 13$.

Section 3: Conic Classification & Discriminants

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7 Classify the conic section represented by $3x^2 + 3y^2 - 12x + 6y - 15 = 0$.

a. **Circle ✓**

b. Ellipse

c. Parabola

d. Hyperbola

Since coefficients of x^2 and y^2 are equal and non-zero ($A = C = 3$) with no xy-term, this represents a circle.

8 Which conic section is represented by the equation $y^2 - 4x - 6y + 1 = 0$?

a. Circle

b. Ellipse

c. **Parabola ✓**

d. Hyperbola

Only one variable (y) is squared while x appears only to the first degree. This is a parabola opening horizontally.

9 What is the discriminant $B^2 - 4AC$ of the general conic $4x^2 - 12xy + 9y^2 + 2x - 5y = 0$, and what does it indicate?

a. 0 **(Parabola) ✓**

b. -48 (Ellipse)

c. 48 (Hyperbola)

d. 0 (Degenerate Circle)

$B = -12, A = 4, C = 9$. $B^2 - 4AC = (-12)^2 - 4(4)(9) = 144 - 144 = 0$. A zero discriminant indicates a parabola.

Section 4: Rotations & Coordinate Shifts

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10 Under a 90° counterclockwise rotation about the origin, what are the new coordinates of the point $(3, -5)$?

a. $(5, 3)$ ✓

b. $(-5, -3)$

c. $(3, 5)$

d. $(-3, 5)$

Rotation formula for 90° counterclockwise: $(x, y) \to (-y, x)$. Here $(3, -5) \to (-(-5), 3) = (5, 3)$.

11 A line $y = 3x - 2$ is reflected across the y-axis. What is the equation of the reflected line?

a. $y = -3x - 2$ ✓

b. $y = -3x + 2$

c. $y = 3x + 2$

d. $y = -\frac{1}{3}x - 2$

Reflection across the y-axis replaces x with $-x$: $y = 3(-x) - 2 = -3x - 2$.

12 What is the distance from the point $(2, 3)$ to the line $3x - 4y + 1 = 0$?

a. 1 ✓

b. 2

c. $\frac{7}{5}$

d. $\frac{5}{7}$

Distance formula: $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|3(2) - 4(3) + 1|}{\sqrt{3^2 + (-4)^2}} = \frac{|6 - 12 + 1|}{5} = \frac{|-5|}{5} = 1$.

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